SafeLink: AI-Powered Mobile Platform Enhancing Public Safety and Community Policing in Sri Lanka

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Final Report

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B.Sc. (Hons) Degree in Information Technology

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DECLARATION OF CANDIDATE AND SUPERVISOR

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

In Sri Lanka, community policing and public safety are dealing with more difficulties. These include slow emergency responses, road accidents, gender-based violence, and the rapid spread of harmful content on social media. Current safety and monitoring methods are mostly manual, reactive, incompatible, and limiting real-time risk detection and proactive intervention. This limits the ability to detect risks in real time and to take proactive measures. This study proposes an Artificial Intelligence (AI)-powered mobile platform to improve public safety by integrating various intelligent subsystems into a single framework. The platform includes four main features: an AI-based module for predicting risk to women and children, an emergency SOS feature with real-time location tracking and live streaming, an AI system to detect harmful social media content, and accident prediction using computer vision and machine learning techniques. By integrating computer vision, natural language processing, deep learning, and predictive analytics into a mobile-friendly design, the suggested system can help identify risks early, send timely alerts, and support data-driven decisions. This solution aims to enhance emergency response, assist in preventive policing, and promote safer designs for the Sri Lankan environment.

Keywords — Public Safety, Community Policing, Emergency SOS, Traffic Violation Detection, Risk Prediction

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LIST OF ABBREVATIONS

AI - Artificial Intelligence

API – Application Programming Interface

CNN – Convolutional Neural Network

GPS – Global Positioning System

IOT – Internet of Things

ML - Machine Learning

NLP – Natural Language Processing

OCR – Optical Character Recognition

REST – Representational State Transfer

SOS – Save Our Souls (Emergency Alert Signal)

UI – User Interface

UX – User Experience

YOLO – You Only Look Once

BERT – Bidirectional Encoder Representations from Transformers

YAMNet – Yet Another Mobile Network (Audio Classification Model)

SDK – Software Development Kit

CI/CD – Continuous Integration / Continuous Deployment

IEEE - The Institute of Electrical and Electronics Engineers

ICAICA - International Conference on Artificial Intelligence and Computer Applications

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1.INTRODUCTION

1.1 Background Study.

Public safety systems have changed a lot in recent years due to the rise of digital technologies and artificial intelligence. However, many current systems, especially in developing countries, still use reactive approaches. These approaches focus on responding to incidents after they happen instead of preventing them. Traditional emergency response methods, like hotline communication systems, rely heavily on voice interaction. This becomes unreliable in high-stress or life-threatening situations. Victims may find it hard to communicate clearly because of panic, injury, or environmental factors. This can cause delays in response and reduce the effectiveness of emergency services [1].

To address these limitations, recent research has focused on using mobile technologies along with real-time data sharing. GPS-based location services help emergency responders quickly find a victim, which reduces response time. However, location information alone does not clearly explain what is happening at the scene. For instance, two alerts from the same location may indicate very different situations, such as a minor issue or a serious life-threatening emergency. Because of this, researchers have recommended combining additional data sources like live video, audio, and other contextual information. Live videos can show what is happening in real time, while audio can capture important sounds like shouting, alarms, or distress signals. When all these data sources are used together, they offer a clearer understanding of the situation. This allows emergency teams to make faster, more accurate, and better-informed decisions [2].

The rise of Artificial Intelligence (AI) has boosted the growth of smart public safety systems. Computer vision methods, especially deep learning models like YOLO, have shown great accuracy in spotting objects such as weapons, fire, and violent behavior in live video feeds. In the same way, audio analysis models like YAMNet and other deep neural networks can identify distress signals, including screams, alarms, and gunshots. These methods, which mix visual and audio data, are more effective than single-modality systems in detecting emergencies and prioritizing responses [3], [4].

In addition, the fast growth of social media platforms has brought new challenges for public safety. Social media is often the first place where reports of incidents like violence, harassment, and accidents appear. However, the huge amount of user-generated content makes it hard to monitor everything by hand. Natural Language Processing (NLP) techniques, including transformer-based models like BERT and XLM-RoBERTa, have proven effective in identifying harmful or abusive content in various languages. When paired with image and video analysis, these models allow for real-time threat detection and help authorities intervene proactively [5].

Road safety is a crucial area where new technologies are used to improve public safety. Traffic violations and road accidents remain significant causes of injuries and deaths worldwide. In many areas, traffic monitoring relies on police officers and basic camera systems. These traditional methods are often ineffective because they require much human effort and can lead to mistakes or delays. Recent research indicates that computer vision technology can automatically detect traffic violations, such as speeding, running red lights, and illegal parking, by analyzing camera footage in real time. Additionally, machine learning models can examine past traffic data to identify patterns and pinpoint locations where accidents frequently occur. By understanding these patterns, the system can predict future risks and alert authorities in advance. This allows for preventive measures, enhances road safety, and lowers the number of accidents [6].

Safety issues concerning women and children have become a significant research focus in recent years. Most safety applications available today react to situations, meaning users must manually send an alert when they are already in danger. This can be tough in real-life situations, as the person may be scared, hurt, or unable to use their phone. To address this, researchers are developing predictive safety systems that can spot risks before incidents occur. These systems use technologies such as location-based analysis and machine learning to examine patterns from past incidents and identify high-risk areas. They can also track how risk changes over time, which helps provide early warnings to users. Many of these systems gather data from the public, like user reports and feedback, to improve their accuracy and reliability. Additionally, multilingual Natural Language Processing is crucial for these systems because people may report incidents in different languages. Supporting multiple languages makes the system more accessible and useful for a broader group of users [7].

Even though many improvements have been made, there is still a significant limitation in current research. Most existing systems focus only on one area, such as emergency response, traffic monitoring, or social media analysis. These systems operate independently and do not connect with each other. Because of this, they cannot share information or collaborate. This lack of integration lowers the overall effectiveness of public safety systems. For example, data from traffic information, social media, and emergency reports could be combined to gain a clearer understanding of a situation, but current systems do not support this. As a result, it becomes hard to get a complete view of risks and make informed decisions. A more connected approach is needed to combine data from various sources and offer a better understanding of public safety issues.

Because of these issues, there is a strong need for one system that can combine various safety features into a single platform. An AI-powered system that connects multiple components can offer a better and more complete solution. By using different types of data such as video, audio, location, and text together, along with real-time monitoring and prediction methods, the system can respond to emergencies more effectively and help prevent problems before they occur.

This integrated approach improves how quickly, and accurately emergency services can respond. It also supports preventive actions by helping authorities identify risky situations early. As a result, they can take the necessary steps before incidents become serious. Overall, developing a united and smart public safety platform is an important step toward creating safer and more secure communities.

1.2 Literature review.

In recent years, many researchers have focused on using artificial intelligence (AI) and real-time data to improve public safety systems. Traditional safety systems, especially in countries like Sri Lanka, mainly rely on manual reporting methods, such as phone calls to emergency hotlines or physical monitoring. These systems are often slow and do not provide enough information about the situation. This makes it difficult for authorities to respond quickly and effectively [8].

In emergency response systems, modern solutions use real-time technologies like GPS tracking, live video, and audio sharing. Systems like Next Generation 112 let users send their location and live data to emergency services. This improves situational awareness compared to traditional voice calls [9]. Research shows that combining video and audio data helps authorities understand emergencies better and respond faster [10]. AI models like YOLO are commonly used to detect objects such as weapons and fire in video streams. Audio models can identify distress sounds, including alarms and shouting [11]. These technologies increase the accuracy and speed of emergency response.

Safety systems for women and children have been studied extensively. Most current applications are reactive, so users must manually trigger alerts. Recent studies have introduced predictive safety systems that use machine learning and location-based data to identify high-risk areas and give early warnings [12]. These systems often rely on clustering and time-based analysis to find patterns in crime data and improve safety planning.

With the rapid growth of social media, harmful content like harassment, violence, and abuse has become a major concern. Manual monitoring is impractical because of the vast amount of data generated every day. Therefore, AI-based systems have been created to automatically detect harmful content. Natural language processing models such as BERT and RoBERTa identify harmful text, while computer vision models analyze images and videos for violent activities [13]. Studies show that combining multiple data types, including text, video, and audio, improves detection accuracy compared to using a single method.

In traffic management, traditional systems depend mostly on manual monitoring and basic CCTV systems. These methods are not effective for large-scale environments. Recent research has introduced smart traffic systems that use AI and computer vision. These systems can identify traffic violations in real time. They also apply machine learning models to analyze historical data and pinpoint areas prone to accidents [14]. This information helps authorities take preventive measures and enhance road safety.

In the realm of real-time learning systems, Sze Ee (2020) conceived a state-of-the-art real-time sign language learning system, illustrating technology's capability to deliver instantaneous feedback and enhance learning outcomes [18].

Most existing systems still operate on their own and focus on just one area, like emergency response, traffic monitoring, or social media analysis. This separation limits how well they work. There is a need for a unified AI-powered platform that brings all these features together into one system to improve prevention and response in public safety.

1.3 Research Gap.

Existing public safety systems in Sri Lanka and many other countries are mostly reactive. They respond only after an incident has occurred. Traditional methods, such as emergency hotlines, manual monitoring, and separate reporting systems, often fail to provide timely and accurate information during critical situations. Victims may struggle to communicate clearly because of stress, danger, or language barriers. This leads to delays and ineffective responses.

Another major limitation identified in current research is that most systems are developed as separate solutions focusing on specific areas like traffic management, social media monitoring, women's safety, or emergency response. These systems work independently and do not share data or insights. As a result, authorities cannot gain a full understanding of public safety situations, which reduces the overall effectiveness of decision-making and response strategies.

Many existing solutions do not process data in real-time and depend mostly on manual input or delayed analysis. This makes it hard to spot incidents as they occur or to act quickly. Additionally, most systems rely on just one type of data, like text or images. They do not use multimodal data, which includes video, audio, and text combined, that can give a better understanding of situations.

Another important gap is the lack of predictive capabilities in existing systems. Many current applications focus only on detecting incidents, not on predicting potential risks. Without predictive analysis, it is hard to identify high-risk areas, prevent crimes, or reduce accidents before they happen.

There is also limited attention on localized solutions for Sri Lanka. Many AI systems are designed for high-resource languages and settings. They do not support bilingual communication in Sinhala and English or take into account local social and cultural factors.

Therefore, there is a clear need for a single, AI-powered public safety platform that combines multiple areas into one system. This system should include real-time monitoring, predictive analytics, and multimodal data processing to improve prevention and response. Filling these gaps will lead to quicker decision-making, better resource allocation, and a more effective way to ensure public safety in Sri Lanka.

Table 1 - Comparison with difference existing Research (Environment Studies Component)

System	**Tool**				
	Research A [16]	**Research B [17]**	**Research C [18]**	**Research D [19]**	**Proposed solution**
Integration of multiple safety domains	✗	✗	✗	✗	✓
Real-time monitoring and alerts	✓	✓	✗	✓	✓
Multimodal data processing (text, video, audio)	✗	✗	✓	✗	✓
Predictive analysis capability	✗	✗	✗	✓	✓
Live streaming support	✓	✗	✗	✗	✓
Bilingual support (Sinhala & English)	✗	✗	✗	✗	✓
Centralized dashboard for authorities	✗	✗	✗	✗	✓
Cross-platform (web + mobile) support	✗	✗	✗	✗	✓

1.4 Research Problem.

Public safety in Sri Lanka faces several challenges because of the limitations of current systems for emergency response, crime detection, and risk prevention. Solutions like emergency hotlines, traffic monitoring systems, social media analysis tools, and women's safety applications mostly work independently and depend a lot on manual processes. These systems are often slow and reactive. They struggle to provide full situational awareness, especially during critical incidents.

One major issue is that traditional emergency communication systems rely on voice-based reporting. Victims must clearly explain their situation. In many real-life scenarios, people may struggle to communicate due to stress, danger, or language barriers. This often results in delayed or incorrect responses. Similarly, traffic monitoring systems only detect violations but do not predict accident risks ahead of time. Additionally, social media monitoring tools are not connected with real-time law enforcement systems, leading to delays in identifying and responding to harmful activities.

Most existing applications also do not support real-time data processing and multimodal analysis. This means they struggle to combine text, video, and audio data to understand complex situations. As a result, the accuracy of detection decreases, and authorities find it harder to make quick, informed decisions. In addition, many systems lack predictive capabilities. This limitation prevents them from identifying high-risk areas or stopping incidents before they happen.

Another critical problem is the lack of a unified platform that combines different safety areas into one system. Without this integration, data stays scattered across several systems. This fragmentation stops authorities from getting a full and clear picture of public safety conditions.

2. OBJECTIVES.

2.1. Main Objective.

The main objective of this research is to design and develop an AI-powered integrated public safety platform that enhances emergency response, crime detection, and risk prediction in Sri Lanka by combining multiple safety components into a single system.

2.2. Specific Objective.

- To develop a Smart Traffic Violation Detection and Accident Prediction system using computer vision and machine learning techniques to identify traffic violations and predict high-risk accident areas.
- To design a Real-Time AI-based Social Media Monitoring system that detects harmful and crime-related content using natural language processing, computer vision, and audio analysis.
- To implement a Safety Prediction system for Women and Children that uses geospatial analysis and machine learning to identify high-risk locations and provide preventive alerts.
- To develop an Emergency SOS system with real-time location tracking and live streaming, enabling users to share video, audio, and GPS data with authorities during emergencies.
- To integrate all components into a unified platform that supports real-time monitoring and data sharing across multiple safety domains.
- To apply multimodal AI techniques (text, video, and audio) to improve the accuracy and reliability of detection and analysis.
- To provide a bilingual (Sinhala and English) user interface to ensure accessibility and usability for a wider range of users in Sri Lanka.

3. METHODOLOGY.

3.1 Requirement Gathering and Analysis

The requirement gathering and analysis phase aimed to identify the key needs and functions of the SafeLink system. This involved reviewing existing public safety systems, including emergency hotlines, traffic monitoring tools, social media detection systems, and women safety applications, to understand their limitations. From this analysis, the main requirements of the system emerged. These included real-time monitoring, accurate incident detection, predictive risk analysis, and the integration of multiple safety components into one platform. We considered both functional requirements, like SOS alerts, traffic violation detection, harmful content identification, and non-functional requirements, such as system performance, security, and scalability. This phase helped define a clear structure for the system and made sure the proposed solution effectively addresses real public safety challenges.

3.1.1 Functional Requirements

a) Real-Time Emergency Alerting:

The system must allow users to send Emergency SOS alerts with real-time GPS location. It should instantly notify the relevant authorities and provide accurate location details during critical situations.

b) Live Streaming Capability:

The system should support live video and audio streaming during emergencies. This allows authorities to understand the situation clearly and respond effectively.

c) Harmful Content Detection:

Students The system must detect harmful and crime-related content from social media using AI models. It should analyze text, images, and videos to identify threats and generate alerts.

d) Traffic Violation Detection:

The system should automatically detect traffic violations using computer vision techniques. It must identify vehicles and capture relevant data such as license plate numbers.

e) Accident Prediction:

The system should analyze historical traffic data to identify accident-prone areas and predict potential risks, helping authorities take preventive actions.

f) Safety Prediction for Women and Children:

The system must identify high-risk areas using location-based analysis and provide safety alerts to users. It should support predictive models to improve personal safety.

g) Real-Time Notifications:

The system should provide real-time alerts and notifications to users and authorities for all detected incidents, ensuring quick response and action.

h) Centralized Dashboard for Authorities:

The system must include an admin dashboard that allows authorities to monitor incidents, view alerts, and manage system activities efficiently.

i) Data Storage and Management:

The system should securely store user data, reports, and incident records, ensure proper data management and easy retrieval when need.

j) Multimodal Data Processing:

The system must process multiple types of data such as text, images, video, and audio to improve detection accuracy and system performance.

k) Bilingual Support:

The system should support Sinhala and English languages to ensure accessibility for a wider range of users in Sri Lanka.

l) System Integration:

The system must integrate all modules (SOS, traffic, social media, and safety prediction) into a single platform to provide a unified and efficient solution.

m) Feedback Mechanism:

The system should include a feedback mechanism where users can report issues or provide suggestions to improve system performance and usability.

3.1.2 Non-Functional Requirements

a) Performance:
The system should process data and generate alerts in real time with minimal delay to ensure quick response during emergencies.

b) Security**:**
The system must protect sensitive user data using secure storage and communication methods such as encryption.

c) Reliability**:**
The system should operate continuously with minimal downtime and provide accurate and consistent results.

d) Usability**:**
The system should have a user-friendly interface that is easy to understand and use for both general users and authorities.

3.2 Feasibility study

3.2.1 Technical Feasibility:

In this research, it is important to assess the technical feasibility. This will help ensure that the proposed SafeLink system can be developed and implemented successfully using available technologies. Several key factors are considered to evaluate the practicality of the system.

a) Data Availability:

   The SafeLink system relies on several types of data, including traffic video data, social media content (text, images, videos), location-based reports, and emergency streaming data. Collecting high-quality and diverse datasets is crucial for training precise AI models. However, getting large-scale, real-world datasets can be tough. Thus, methods like data augmentation, publicly available datasets, and simulated data generation can be used to improve model performance.

b) Hardware Resources:

   High-end The development of AI models for object detection, natural language processing, and prediction requires high-performance computing resources. Training deep learning models may require GPUs or cloud-based computing platforms. Additionally, for real-time system deployment, scalable cloud infrastructure is required to handle large volumes of data and user requests efficiently.

c) Software Resources:

The SafeLink system will be developed with modern software technologies. The mobile application will be built using Flutter to support development across different platforms. The backend will use FastAPI to manage system operations and API communication. We will create AI models with Python-based frameworks like TensorFlow and PyTorch, along with OpenCV for computer vision tasks. For social media analysis, we will use NLP models such as BERT or XLM-R. Additionally, we will employ version control and continuous integration tools like GitHub and GitHub Actions to maintain and streamline the development process.

3.2.2 Financial Feasibility

The development and implementation of the SafeLink system involve several costs. Most software technologies used in the system, such as Python, React Native and AI frameworks, are open source and can be used without licensing fees. As a result, software costs are minimal. The main expenses come from cloud computing services needed for data processing, storage, and real-time system operations. Additionally, there may be costs for hosting servers, maintaining databases, and deploying the system. If the mobile application is published, registration fees for platforms like the Google Play Store and Apple App Store must also be taken into account. Overall, the system is financially feasible since it relies mainly on open-source technologies and scalable cloud resources.

3.2.3 Scheduling Feasibility

The SafeLink project is set to be finished within a year. To finish successfully, we need a clear project schedule. The development process breaks down into several phases, such as gathering requirements, designing the system, developing the model, creating the application, integrating, testing, and deploying. Each phase has specific milestones and deadlines to monitor progress. A detailed project plan serves as a guideline, making sure that all parts are completed on time and that resources are managed well during the development process.

3.3 Data Collection and Preparation:

The SafeLink system needs a variety of data from different sources. This includes traffic video data, social media content like text, images, and videos, user reports, and location information. The data comes from publicly available datasets, APIs, and simulated inputs when needed.

Once collected, the data goes through preprocessing to make sure it is consistent, accurate, and high quality. Preprocessing steps include cleaning, normalizing, annotating, and formatting the data according to the needs of various AI models. For image and video data, we may use augmentation techniques to improve model performance. We process text data using natural language processing techniques, and we organize location data for geospatial analysis. Proper data preparation helps the system train models effectively and makes accurate predictions along with real-time analysis.

Table 2 - Dataset Composition

Table 3 - Dataset Composition

Table 4 - Dataset Composition

Table 5 - Dataset Composition

3.4 Model Development.

Deep learning and machine learning techniques are used to develop models for various parts of the SafeLink system. Computer vision models like YOLO (You Only Look Once) help with object detection in traffic monitoring and emergency video analysis. Natural Language Processing (NLP) models such as BERT and XLM-R analyze social media text data to find harmful content. Audio processing models identify distress sounds from live streaming data. Additionally, machine learning algorithms like clustering techniques, for example, DBSCAN, and predictive models assist with safety risk prediction and accident analysis.

The models are trained with the prepared datasets gathered from various sources. During training, we adjust hyperparameters and improve performance through repeated testing. We use model evaluation techniques to guarantee accuracy, reliability, and efficiency in real-time situations. We constantly improve the models to increase detection accuracy and the system's predictive abilities.

3.5 Front-End Component.

The front-end component of the SafeLink system is primarily developed using the The front-end part of the SafeLink system is mainly built with the Flutter framework. Flutter was chosen because it supports cross-platform development, which lets the app run on both Android and iOS devices with one codebase. Unlike complicated graphical apps, the SafeLink system doesn’t need advanced 3D rendering or game-like interactions. This makes Flutter a good and effective choice for creating a responsive and user-friendly interface.

The main function of the front-end application is to let users interact with the system in real time. Users can trigger Emergency SOS alerts, share their GPS location, and start live video and audio streaming from their mobile device. The application also offers features to view safety alerts, risk predictions, and notifications generated by the system. In addition, users can report incidents and get updates about traffic violations, harmful social media content, and safety warnings.

The mobile application connects to the backend server through APIs. Data like video streams, images, text inputs, and location details are sent for processing. After the backend processes the data with AI models, it sends the results back to the front-end application. The application then shows alerts, predictions, or responses to the user clearly and understandably.

The interface is designed to be simple, intuitive, and easy to use. This ensures it is accessible to a wide range of users. It supports bilingual functionality in Sinhala and English to make it more user-friendly in Sri Lanka. Overall, the front-end component serves as the main interaction layer between users and the SafeLink system. It allows real-time communication and effective responses during safety-related situations.

3.6 User Interface design

The SafeLink system's user interface is built to meet the identified needs. It aims to be easy to use, clear, and efficient for users in safety-related situations. The interface highlights quick access to essential features, including Emergency SOS, safety alerts, and real-time notifications. It keeps a simple and intuitive design to minimize confusion during emergencies.

The user interface is created with custom-built Flutter components. These components are then deployed as native interfaces for Android and iOS. The design keeps consistency across devices and offers a responsive experience for users. The layout is organized in a way that allows users to navigate between features easily.

The SafeLink application has four main components: Emergency SOS, Women and Children Safety Prediction, Harmful Social Media Content Detection, and Smart Traffic Violation Detection. Each component appears as a separate module in the application, so users can access specific features depending on their needs. The Emergency SOS module allows for quick access, letting users send alerts and start live streaming with just one action. The safety prediction module shows risk levels and alerts based on the user's location. The social media detection module displays harmful content that has been identified and related alerts. The traffic module provides information on violations and areas where accidents are likely to occur.

The interface features notification panels, alert messages, and visual indicators to help users clearly understand system outputs. We focus on usability by supporting bilingual functionality in Sinhala and English to make it accessible for more users. The design emphasizes clarity, fewer interaction steps, and quick responses, which are crucial for managing real-time public safety situations.

The following figures show the user interfaces that have been created. They also demonstrate how various modules and features are arranged to deliver a smooth and effective user experience.

3.7 Back-end component

The back-end part of the SafeLink system runs in a server environment. It processes, analyzes, and manages data from various system modules. This component serves as the main link between the front-end application, AI models, and databases. The back end is built with the Python FastAPI framework, which creates efficient and scalable REST APIs for user requests, real-time data transmission, and system integration.

The back end receives data from the front-end application, including video streams, images, text inputs, and GPS location data. Different AI models, such as YOLO for object detection, NLP models for text analysis, and audio processing models for sound detection, process these inputs. The back end then stores the processed results in the database and sends them back to the user interface or administrative dashboard as alerts, predictions, or notifications.

For system efficiency and scalability, the backend architecture is designed using two approaches as follows:

follows.

- Pattern 1: Centralized Processing Architecture.

  In this approach, all incoming data from different modules (SOS, traffic, social media, and safety prediction) is processed through one centralized system. This method makes the structure simpler and allows for easier data management. However, it can lead to longer processing times and lower performance when handling large amounts of data at the same time.

- Pattern 2: Distributed Multi-Model Architecture

  To improve performance and scalability, a distributed architecture is introduced. In this setup, different AI models are deployed separately for each module, like traffic detection, social media analysis, and risk prediction. An API gateway routes incoming data to the right model based on the request type. This method reduces processing time, improves accuracy, and boosts system efficiency by handling tasks in parallel.

The back end also connects with cloud-based databases like Firebase and PostgreSQL to store user data, incident reports, and system outputs securely. We use proper authentication and security measures to protect data privacy and ensure safe communication between system components.

3.8 Integration with Game Framework:

The developed modules are combined into a single system framework that supports real-time public safety operations. This integration links the front-end mobile application, backend server, AI models, and database systems to guarantee smooth communication among all components. Each module, such as Emergency SOS, Social Media Detection, Safety Prediction, and Traffic Monitoring, communicates with the centralized backend using REST APIs.

User interactions like sending SOS alerts, reporting incidents, or accessing safety information go through the integrated system. The backend directs incoming data to the appropriate AI models, processes the information, and produces outputs like alerts, predictions, and notifications. The results are then shown to users and authorities via the mobile application and administrative dashboard, making sure the system works effectively and in coordination.

3.9 User Testing and Feedback.

Usability testing involves a group of target users, which includes general mobile users and system administrators. The goal is to assess the functionality and usability of the SafeLink system. The testing focuses on how responsive the system is, how easy it is to use, and how effective the safety features are, like SOS alerts, notifications, and predictions.

Feedback is gathered through user interactions, surveys, and observations. We use this feedback to make improvements in user experience, interface design, and system performance. This ongoing process keeps the system user-friendly, reliable, and appropriate for real-world public safety applications.

3.10 Evaluation and Validation.

Based on the evaluation results and user feedback, the system is improved for better performance, accuracy, and usability. We address the identified issues and make the necessary changes to optimize system functionality.

Qualitative evaluation is done by collecting user feedback, conducting surveys, and observing system usage. This helps us understand how well the platform works and how easy it is to use. This validation process makes sure that the system meets user expectations and tackles public safety challenges effectively.

3.11 Refinement and Deployment.

Based on the evaluation results and user feedback, the system is improved to boost performance, accuracy, and usability. We address identified issues and carry out necessary improvements to optimize system functionality.

The final version of the SafeLink system is deployed using cloud-based infrastructure to ensure scalability, availability, and real-time performance. The system is accessible through mobile and web platforms, allowing users and authorities to use its features effectively. Continuous monitoring and updates ensure system reliability and respond to changing public safety needs.

3.12 System Overview Diagram.

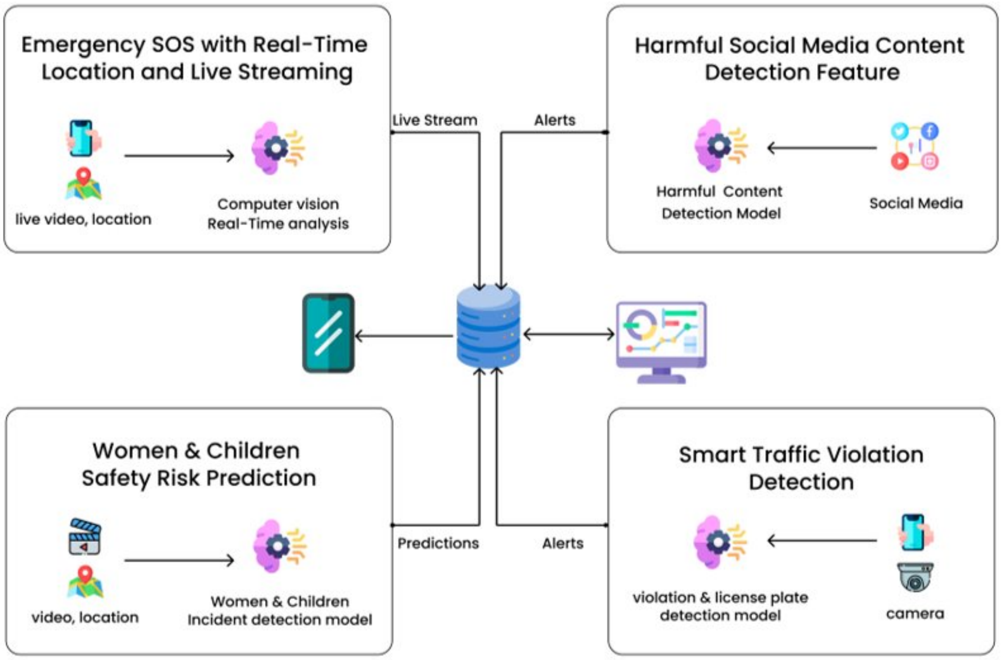

Figure 1 - System Overview Diagram

3.12 System Description.

Figure 1 shows the overall system structure of the proposed SafeLink platform. It is an AI-powered public safety system. The system includes four main functional modules: Emergency SOS with Real-Time Location and Live Streaming, Harmful Social Media Content Detection, Women and Children Safety Risk Prediction, and Smart Traffic Violation Detection. These modules connect through a centralized backend server and database, which serve as the system's foundation.

The Emergency SOS module allows users to send real-time alerts with their GPS location and live video or audio streams. The system analyzes the incoming data with computer vision techniques to spot possible threats in real time. The processed information is sent to the central server, where alerts are created and sent to the appropriate authorities.

The Harmful Social Media Content Detection module gathers data from social media platforms and processes it with artificial intelligence models. This module finds harmful or crime-related content in text, images, and videos. After detecting such content, it generates alerts and sends them to the central system for monitoring and action.

Mobile The Women and Children Safety Risk Prediction module focuses on identifying unsafe areas and predicting potential risks. It uses location data and AI-based detection models to analyze patterns and generate safety predictions. These predictions go to the central system to support preventive decision-making and improve public safety.

The Smart Traffic Violation Detection module processes data from traffic cameras to identify violations and evaluate accident risks. AI detection models are used to find traffic-related incidents. The results are sent as alerts and predictions to the central server.

All modules connect to the central backend system through secure data channels. The backend processes incoming data, creates alerts and predictions, and stores important information in the database. The final outputs appear on an administrative dashboard. This lets authorities monitor real-time incidents, examine trends, and take appropriate actions.

Overall, the system brings together several safety areas into one platform. It allows for real-time monitoring, predictive analysis, and effective communication between users and authorities. This unified approach improves situational awareness and makes public safety management more effective.

3.13.1 Development Process.

3.13.1.1 Emergency SOS with Real-Time Location & Live Streaming Feature

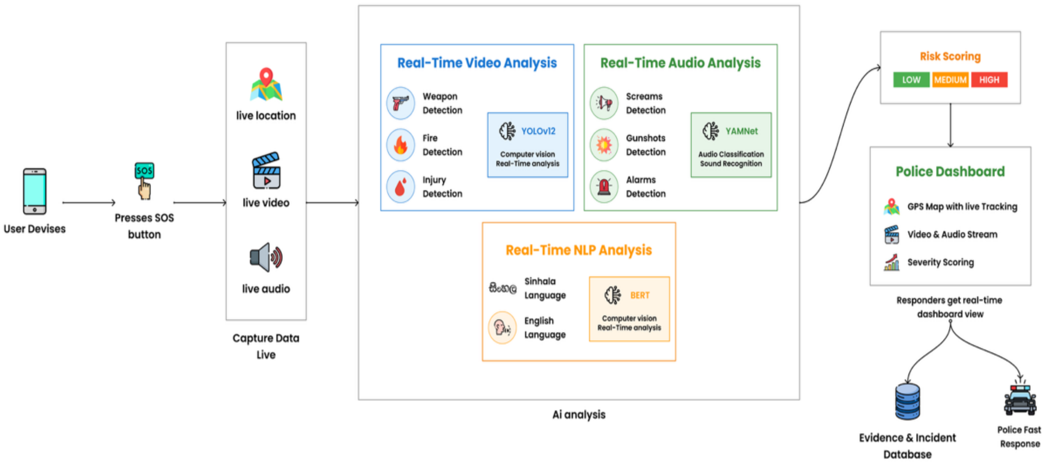

Figure 2 - Component diagram for Emergency SOS

1. Front-End Component:

   The front end of the Emergency SOS component is built with the Flutter framework for cross-platform mobile app development. The user interface is designed to be simple and easy to access in emergency situations.

   The development process starts with adding the SOS trigger feature. Users can activate the emergency alert by pressing one button. Once activated, the application captures the GPS location in real time using the device's location services. At the same time, the app begins live video and audio streaming using a real-time communication SDK like Agora.

   The captured data, which includes location, video, and audio streams, is sent to the backend server through secure API calls. The interface also gives visual feedback to the user, like alert confirmation and streaming status. We pay special attention to usability by keeping interaction steps minimal, ensuring a fast response, and providing bilingual support in Sinhala and English.

2. Back-End Component:

The backend component is built with FastAPI to manage real-time data processing, communication, and integration with AI models. It serves as the main system that gets and processes incoming data from the mobile application.

When an SOS alert is triggered, the backend gets GPS location, live video frames, and audio streams. The system processes video data using computer vision models like YOLO to identify objects such as weapons, fire, or injuries. It analyzes the audio data with sound classification models like YAMNet to detect distress signals such as screams, gunshots, or alarms.

Additionally, NLP models like BERT are used when any text-based inputs are included. The outputs from these models are combined to create a risk score: low, medium, or high. This information is then sent to the police dashboard in real time, along with live tracking and streaming data.

The backend stores incident data, video snapshots, and alerts in a database for future analysis and evidence. API gateways and routing methods help manage multiple requests and ensure the system can grow.

3. Question Bank:

   In the context of the Emergency SOS component, the "Question Bank" is the collection of predefined detection rules, AI model classes, and training datasets used to identify emergency situations.

   The development process includes defining important categories of emergency situations like weapon detection, fire detection, injury detection, and distress sound detection. For each category, we collect and prepare relevant datasets. This includes image datasets for detecting objects and audio datasets for classifying sounds.

   The datasets go through cleaning, annotation, and augmentation techniques to improve model performance. AI models are then trained to recognize specific patterns and classify inputs accurately. For example, the YOLO model is trained on labeled images of weapons and fire. The YAMNet model is trained in different types of emergency sounds.

In addition to training the models, we set detection thresholds and rules to decide when to trigger an alert. These rules help reduce false positives and improve the system's reliability. We combine the outputs from different models to create a final risk score. This score aids in decision-making and generating alerts.

3.13.1.2 AI-Powered Women and Children Safety Risk Prediction Feature

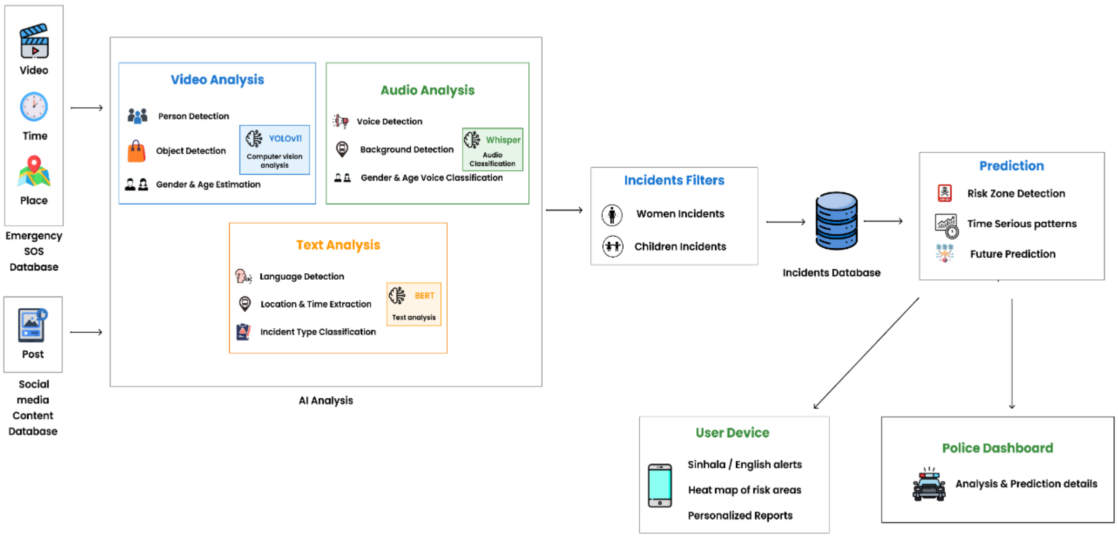

Figure 3 - Component diagram for Safety Prediction System Feature

1. Data Collection and Preprocessing:

   The system gathers incident-related data from various sources, including video recordings, audio segments, and written reports about the safety of women and children. This data comes from publicly available datasets and simulated inputs when needed. After collecting the data undergo preprocessing to ensure quality and consistency. This involves normalizing videos, cleaning audio, and standardizing Sinhala-English text. These steps are crucial for preparing the data for precise analysis across different data types.

2. Multimodal Data Analysis:

   The system examines the collected data with various AI methods. It processes video data with a YOLO-based object detection model to spot women, children, and important contextual elements. The system transforms audio data into text using a speech recognition model, which helps it capture verbal signs of unsafe situations. It analyzes textual data, including transcripts and reports, with a multilingual NLP model like XLM-RoBERTa to sort incidents into categories such as harassment, unsafe environments, or normal activity.

3. Risk Prediction and Pattern Analysis:

   The system combines outputs from video, audio, and text analysis using a multimodal fusion approach to improve reliability. It stores the processed data with metadata like location and time, which helps identify spatial and temporal patterns. Using these patterns, the system predicts safety risk levels for specific areas and times, categorizing them as low, medium, or high risk.

4. Alert Generation and System Integration:

   Based on the predicted risk levels, the system generates alerts and safety notifications for users. These alerts show up in the mobile application with maps, heat maps, and visual indicators to help users easily understand risk levels. The backend system takes care of data processing and communication between components. This ensures real-time predictions and efficient system performance.

5. Testing, Evaluation, and Deployment:

   The system is tested in various scenarios to check its accuracy, reliability, and usability. We use performance metrics like precision, recall, and F1-score to measure how effective the model is. After validation, we deploy the system on cloud-based infrastructure for better scalability and real-time performance. We continually apply updates to enhance prediction accuracy over time.

The AI-powered Women and Children Safety Risk Prediction system offers an effective way to identify potential safety risks through multimodal data analysis. It combines video, audio, and text processing with predictive modeling. This setup allows for early detection of unsafe situations and supports proactive decision-making. The results show that using AI techniques with real-time data can greatly improve risk awareness and personal safety. Overall, the system helps create a smarter and more preventive public safety solution.

3.13.1.3 Real-Time AI System for Detecting Harmful Social Media Content Feature

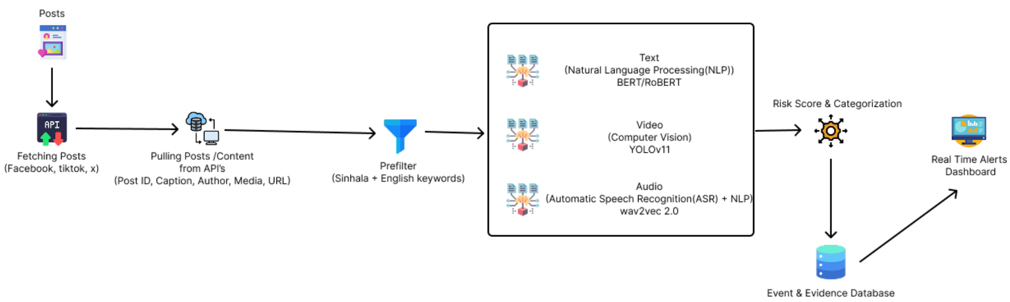

Figure 4 - Component diagram for Real-Time Harmful Social Media Content Detection Feature

1. Requirement Analysis and System Design:

   The development of the Real-Time Harmful Social Media Content Detection component starts with a thorough requirement analysis. This phase involves identifying system goals, defining functional and non-functional requirements, and understanding harmful content on social media platforms. The system aims to detect harmful content such as hate speech, harassment, violent threats, and abusive language across various data types including text, images, videos, and audio. Special attention is given to supporting Sinhala-English code-mixed language, which is often used in Sri Lankan social media. Key requirements include real-time processing ability, high detection accuracy, scalability, efficient data handling, minimal latency, and a user-friendly interface for clear visualization of results.

2. Data Collection and Preprocessing:
   Data from social media is collected using platform APIs and streaming sources, including posts, comments, images, and videos. The collected data goes through preprocessing to ensure quality and consistency. Text data is cleaned, normalized, and tokenized by removing irrelevant elements such as stop words and special characters. Image and video data are resized and normalized. Audio data is extracted from video streams and prepared for transcription. These preprocessing steps are essential for improving model performance and ensuring accurate analysis across different data types.

3. Model Development and Multimodal Analysis:
   The system uses multiple Artificial Intelligence models to analyze different data types. For text analysis, the XLM-RoBERTa model is chosen due to its ability to handle multilingual and code-mixed language. For visual analysis, a YOLO-based object detection model detects harmful elements like weapons, fire, blood, and violent scenes. For audio processing, Automatic Speech Recognition models like Wav2Vec2 convert speech into text, which is then analyzed using the NLP model. This multimodal analysis ensures thorough detection of harmful content across text, image, and audio formats.

4. System Integration and Real-Time Processing:
   The outputs from text, image, and audio models are combined using a multimodal fusion mechanism to enhance detection accuracy. A score-based approach aggregates confidence scores from each type and generates an overall risk score. The system is integrated into a unified pipeline that processes incoming data in real time. The backend, developed with Python and FastAPI, manages data processing, model inference, and communication between components. The results are shown through a real-time dashboard that presents detected categories and confidence scores, making it easy to interpret harmful content.

5. Deployment, Testing, and User Interface Development:

The frontend of the system is developed using React.js to create an interactive and responsive user interface. The dashboard allows users to monitor detection results and analyze harmful content in real time. The system undergoes testing with various datasets and real-world scenarios to evaluate performance using metrics like precision, recall, and F1-score. Finally, the system is deployed on cloud-based infrastructure to ensure scalability, reliability, and continuous operation. Security measures and access controls are put in place to protect data and ensure safe system use.

The development process takes a structured and integrated approach, combining several technologies and methods to create a strong real-time harmful content detection system. By utilizing multilingual processing, multimodal analysis, and real-time monitoring, the system offers an effective solution for improving online safety and enabling early detection of harmful activities.

3.13.1.4 Smart Traffic Violation Detection & Accident Prediction Feature

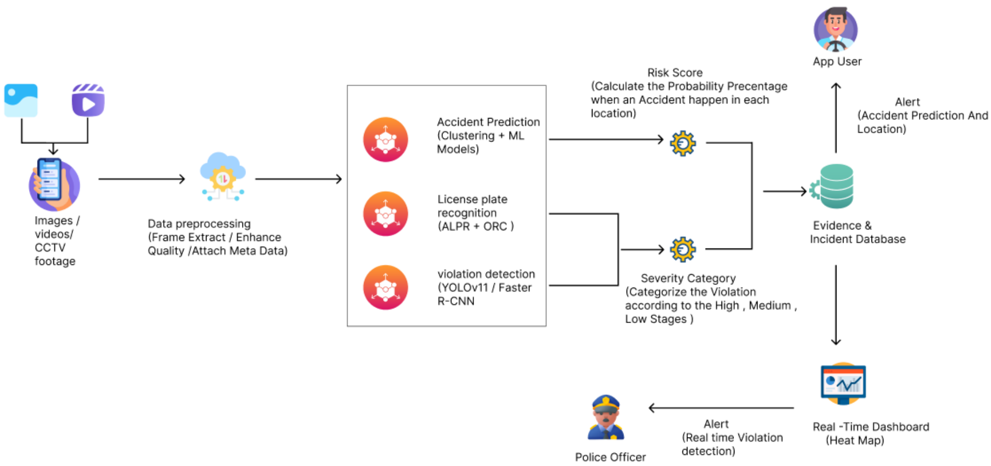

Figure 5 - Component diagram for General Skills component.

1. Data Collection and Preprocessing:

The system collects traffic-related data from various sources. This includes CCTV camera feeds and user-uploaded images or videos through the SafeLink application. The dataset features different vehicle categories and number plate annotations captured under various road and environmental conditions. After collection, the data goes through preprocessing to ensure quality and consistency. This step involves resizing images, normalizing them, and augmenting data to handle variations like lighting changes, motion blur, and occlusions. These steps are crucial for improving the reliability and accuracy of the detection models.

2. Traffic Violation Detection:
   The system analyzes visual traffic data with a YOLOv11-based object detection model. This model detects vehicles such as cars, buses, vans, lorries, bikes, and three-wheelers, along with number plates in real time. Each detected object receives a class label and a confidence score. Non-maximum suppression removes duplicate detections. Based on spatial relationships and predefined rules, the system identifies possible traffic violations like improper lane usage and unsafe vehicle positioning.

3. License Plate Recognition:
   Once vehicles are detected, the system extracts the number plate regions and processes them with PaddleOCR. The OCR module converts the image-based plate information into readable alphanumeric text. This allows for automatic identification of vehicles involved in violations, aiding enforcement and record-keeping.Testing, Evaluation, and Deployment:

4. Accident Prediction and Pattern Analysis:
   The system also analyzes traffic conditions using spatial and contextual information such as vehicle density and positioning. By observing these patterns, it estimates accident risk levels for specific scenarios. The prediction output is categorized into different risk levels, enabling proactive identification of potentially hazardous situations on the road.

5. Alert Generation and System Integration:

Based on detected violations and predicted risk levels, the system generates alerts and visual outputs through a monitoring dashboard and mobile application. Users and authorities receive notifications along with supporting evidence like detected images and vehicle details. The backend system ensures a smooth data flow between detection, OCR, and visualization components, allowing near real-time system performance.

6. Testing, Evaluation, and Deployment:
   The system undergoes testing with various traffic scenarios to assess detection accuracy and reliability. Performance is measured using metrics like precision, recall, F1-score, and mean Average Precision (mAP). After validation, the system is deployed in a scalable environment to support real-time monitoring. Continuous improvements are made to enhance detection accuracy and prediction capability.

The proposed Smart Traffic Violation Detection and Accident Prediction system offers an effective solution for automated traffic monitoring. By combining object detection, license plate recognition, and predictive analysis, the system helps identify violations and potential accident risks early. This approach improves road safety and supports proactive decision-making, contributing to a smarter and more efficient traffic management system.

3.5. Software Development Life Cycle (SDLC).

The Agile SCRUM methodology was chosen for this research project to ensure an efficient, flexible, and ongoing development process. This approach supports continuous improvement and lets the team respond to changing requirements during the project lifecycle. Here's why:

Flexibility for Evolving Requirements: Agile SCRUM allows the team to handle changing requirements without interrupting the overall development process. Its flexible structure lets new features and improvements fit in smoothly based on project needs and feedback.

Iterative Delivery for Continuous Progress: The project is developed through several iterations, or sprints. Tasks are broken down into smaller, manageable parts. At the end of each sprint, a functional piece of the system is delivered. This approach allows for early validation and ongoing progress.

Transparency and Prioritization: Agile SCRUM offers clear visibility of project tasks by using sprint planning and managing the backlog. This lets the team focus on the most important features first and ensures that high-value functions get developed early.

Team Collaboration and Role Management: With a team of four members, Agile SCRUM supports effective collaboration and clear role distribution. Each member works on specific components while collaborating to achieve overall system integration. Supervisors give guidance and feedback throughout the development process.

Effective Communication and Awareness: Regular SCRUM activities like sprint meetings, reviews, and discussions keep team members in touch. This helps spot challenges early and make needed changes to keep the project on track.

Overall, Agile SCRUM methodology fits the project requirements by supporting flexibility, continuous development, effective collaboration, and structured communication. These benefits make it a good approach for managing the development of this research project.

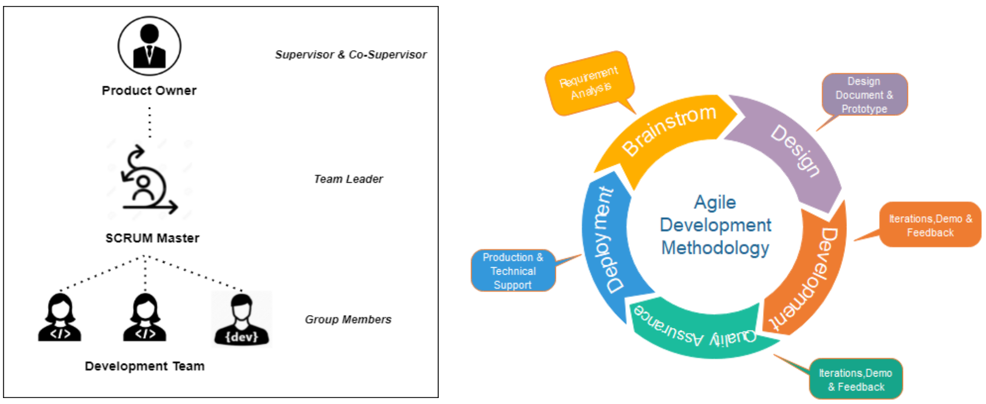

Figure 6 - SCRUM Hierarchy and Agile Methodology.

3.6. Gantt Chart.

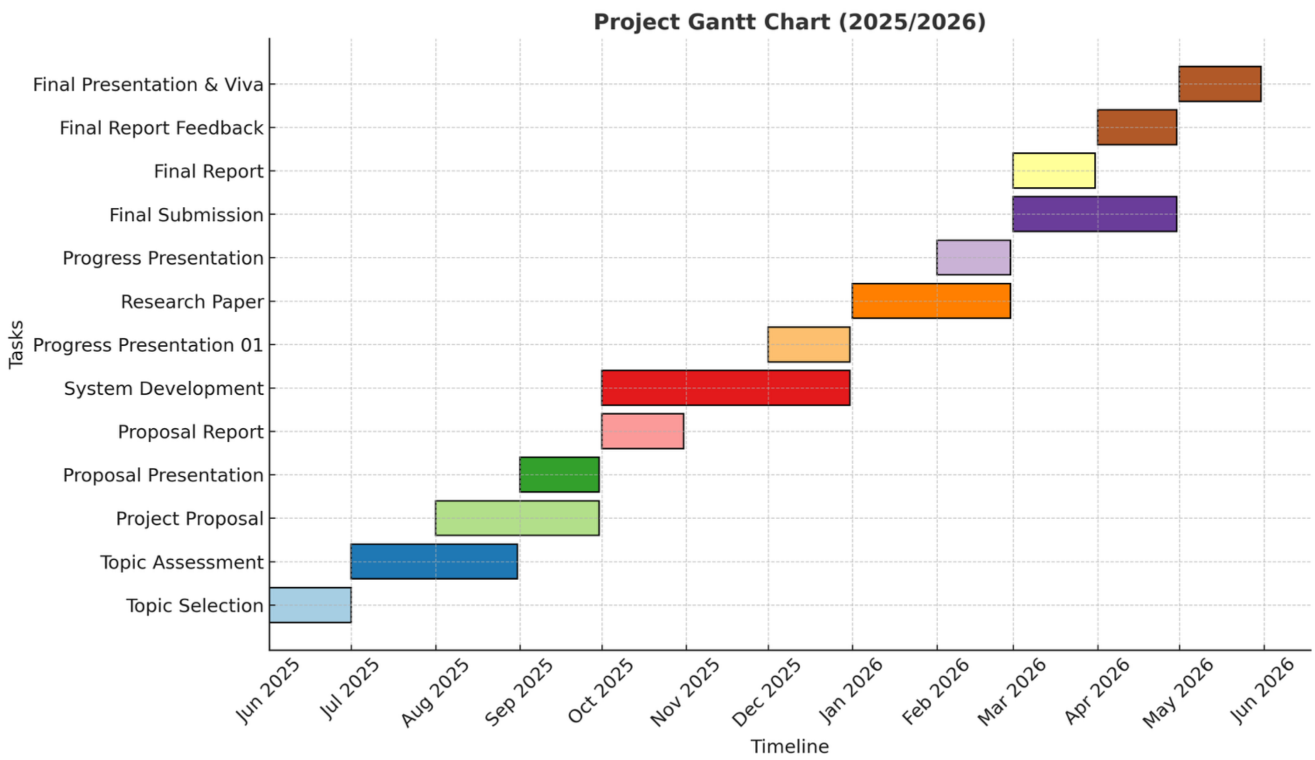

Figure 7 - Gantt Chart.

3.7. Work Breakdown Structure.

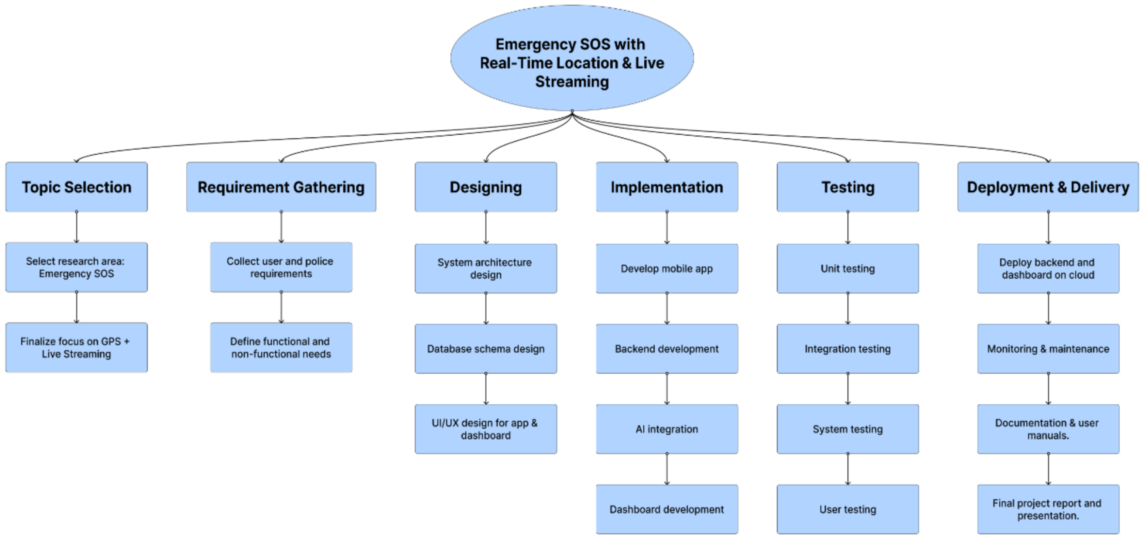

Figure 8 - Work Breakdown Structure.

3.8. Commercialization Aspects of the Products

To effectively commercialize our mobile app tailored for hearing-impaired primary school students, we've devised a comprehensive strategy that focuses on providing value to users while ensuring sustainable revenue streams. Here's our approach:

1. Freemium Model: Our app will be available for free on app stores, offering basic features and access to select study modules. Users can explore these features without any initial cost, allowing them to experience the app's functionality firsthand.
2. Premium Features: For users interested in unlocking advanced functionalities like additional study modules and personalized learning paths, we offer premium subscription options. This tiered approach enables users to evaluate the app's value before committing to a subscription.
3. Subscriptions: We'll provide subscription plans catering to users seeking access to premium features and exclusive content. Users can choose from flexible subscription options, including monthly, quarterly, or annual plans, based on their preferences and budget.

4. Premium Benefits: Subscribers will enjoy benefits such as ad-free browsing, unlimited access to all study modules, progress tracking, personalized learning recommendations, and priority customer support. These features enhance the user experience and encourage subscription uptake.
5. Advertising: To diversify revenue streams, our app will incorporate targeted advertising through platforms like AdMob. Third-party advertisements will be strategically placed within the app to ensure relevance and minimal disruption to the user experience.
6. Revenue Generation: We'll generate revenue through various advertising models such as pay-per-click (PPC), pay-per-impression (PPI), or pay-per-action (PPA). By maximizing user engagement and ad visibility, we aim to optimize advertising revenue while maintaining user satisfaction.
7. In-App Purchases: Additionally, we'll offer in-app purchases for users interested in accessing premium content or features on a one-time basis. These purchases may include exclusive study materials, virtual rewards, or customization options, providing users with additional value and personalization opportunities.

By implementing a multifaceted commercialization strategy that includes the freemium model, subscriptions, advertising, and in-app purchases, we aim to balance accessibility, value, and revenue generation. This approach ensures that our mobile app remains financially sustainable while delivering enriching educational experiences to our target audience of hearing-impaired students.

1. RESULTS & DISCUSSION

4.1 Results

4.1.1 Emergency SOS with Real-Time Location & Live Streaming Feature

The Emergency SOS system was successfully developed and tested to evaluate its performance in real-time emergency scenarios. The system demonstrated the ability to provide instant communication between users and emergency responders through real-time GPS tracking and live audio/video streaming.

During testing, the one-tap SOS feature triggered alerts immediately with minimal delay. The GPS tracking functionality accurately transmitted user location data, allowing continuous monitoring during active sessions. This ensured that responders could effectively track user movement.

The live streaming component functioned with low latency, enabling real-time transmission of video and audio data from the user's device to the backend system. This provided clear situational awareness, allowing responders to observe emergency conditions before arriving at the location.

The AI-based threat detection models produced promising results. The computer vision model was able to detect critical objects such as fire, weapons, and suspicious activities with satisfactory accuracy under controlled testing conditions. Similarly, the audio detection model successfully identified distress-related sounds such as shouting and alarms. These detections were processed in near real-time and integrated into the alert system.

The risk-based prioritization mechanism also showed effective performance by assigning higher priority to alerts originating from high-risk locations or those containing detected threats. This helped distinguish critical emergencies from less serious situations.

Overall, the system achieved its primary objective of providing a real-time, intelligent emergency response mechanism with improved situational awareness and faster alert processing.

4.1.2 AI-Powered Women and Children Safety Risk Prediction Feature

Our The results of the Women and Children Safety Risk Prediction System demonstrate the effectiveness of using multimodal data analysis for identifying and predicting safety risks. The system was tested using a combination of textual, audio, video, and location-based data to evaluate its performance in detecting incidents and generating risk predictions.

The developed models were able to successfully classify safety-related incidents into relevant categories such as harassment, unsafe environments, and normal conditions. The text-based model showed reliable performance in identifying harmful or risk-related content from bilingual inputs, while the video analysis model was able to detect potentially unsafe situations by recognizing contextual visual patterns. Similarly, the audio-based model effectively identified distress signals such as shouting or panic-related sounds, contributing to the overall detection process.

The integration of these models using a multimodal approach improved the overall system performance. Compared to individual models, the combined system provided more accurate and consistent predictions by analyzing multiple sources of data simultaneously. This integration reduced the chances of false detection and increased the reliability of the results.

In addition to detection, the system was able to generate risk predictions based on location and time patterns. High-risk areas were identified using historical and processed data, and these were visualized through heatmaps. The heatmap results clearly highlighted zones with higher safety risks, allowing users to easily interpret the information and take precautionary measures.

The system also demonstrated efficient performance in delivering real-time alerts. Users were able to receive notifications about potential risks based on their location, and the response time of the system was within an acceptable range for practical use. The mobile application successfully displayed alerts, risk levels, and safety insights in a clear and user-friendly manner.

Overall, the results indicate that the proposed system performs effectively in detecting incidents, predicting risks, and presenting safety information. The combination of artificial intelligence techniques and user-centered design contributes to a reliable and practical solution for enhancing the safety of women and children.

4.1.3 Real-Time AI System for Detecting Harmful Social Media Content Feature

The development and deployment of our mobile application aimed at facilitating environmental studies for hearing-impaired primary school students have yielded promising results. Through rigorous testing and evaluation, we have observed several key outcomes and performance metrics, demonstrating the effectiveness and impact of our system.

1. User Engagement and Satisfaction:

   Feedback from user testing sessions and surveys indicates high levels of user engagement and satisfaction with the application interface, content, and interactive features. Hearing-impaired students expressed enthusiasm and interest in using the app as a supplementary learning tool, highlighting its effectiveness in capturing their attention and facilitating comprehension of environmental concepts.

2. Sign Language Recognition Accuracy:

   The sign language recognition module, powered by machine learning algorithms, demonstrated commendable accuracy in interpreting user gestures and translating them into actionable commands within the application. Through extensive training and fine-tuning, the model achieved high levels of precision and recall, enabling seamless interaction with the app's functionalities.

3. Learning Outcomes and Knowledge Retention:

   Preliminary assessments conducted among users revealed significant improvements in environmental literacy and knowledge retention following regular use of the application. Students demonstrated enhanced comprehension of environmental concepts, vocabulary acquisition, and retention of learning outcomes, indicating the efficacy of the app as an educational aid.

4. Accessibility and Inclusivity:

The inclusive design of the mobile application, tailored to the specific needs of hearing-impaired learners, has contributed to fostering a supportive and accessible learning environment. By providing sign language interpretation, visual aids, and interactive exercises, the app has effectively bridged communication barriers and empowered students to actively engage with educational content.

5. Scalability and Future Enhancements:

   The modular architecture of the application, coupled with cloud-based infrastructure, ensures scalability and flexibility to accommodate future updates, expansions, and enhancements. As user feedback and technological advancements emerge, we remain committed to iteratively improving the app's features, performance, and accessibility to meet the evolving needs of our target audience.

6. Stakeholder Feedback and Endorsements:

   Positive feedback and endorsements from educators, parents, and educational institutions underscore the value proposition and societal impact of our mobile application. Stakeholders recognize the potential of the app to revolutionize environmental education for hearing-impaired students and advocate for its widespread adoption in school curricula.

In summary, the overall system results reflect the successful development and implementation of a mobile application that effectively addresses the educational needs of hearing-impaired primary school students. By leveraging technology, inclusive design principles, and user-centered methodologies, we have created a transformative learning tool that empowers students to thrive academically and environmentally.

4.1.4 Smart Traffic Violation Detection & Accident Prediction Feature

The proposed Smart Traffic Violation Detection and Accident Prediction system worked well in meeting its goals, as shown through testing and evaluation. The object detection model accurately recognized various vehicle types and license plates, allowing for reliable identification in different traffic situations. It consistently detected objects under different lighting, angles, and backgrounds, showing strong generalization.

Additionally, using PaddleOCR for license plate recognition allowed for accurate vehicle registration number extraction, supporting automated identification and enforcement. The system could detect and process multiple vehicles in one frame, making it suitable for real-time traffic monitoring. the use of rule-based logic helped the system spot potential traffic violations by analyzing relationships and positions of detected objects. The accident prediction part improved the system by looking at traffic patterns like vehicle density and spatial layout, helping estimate risk levels in various traffic conditions.

Lastly, the system offered clear visual outputs with bounding boxes, confidence scores, and detected labels, making it easier for users and authorities to understand. Ongoing testing and validation confirmed stable performance, showing that the system is a practical option for automated traffic monitoring and improving road safety.

4.2 Research Findings

The implementation and evaluation of the SafeLink system revealed several important findings about how effective it is to use AI technologies for public safety.

First, the results show that analyzing multiple types of data greatly improves detection accuracy and reliability. The combination of video, audio, and text processing in the women and children safety module produced more consistent and meaningful results compared to using just one type of data. This confirms that combining different data sources improves system performance in complex real-world situations.

Second, the study found that real-time communication features greatly improve emergency response efficiency. The Emergency SOS module demonstrated that combining GPS tracking with live audio and video streaming provides better situational awareness than traditional voice-based systems. This helps responders make faster and better decisions during critical situations.

Another key finding is that AI-based automation reduces reliance on manual monitoring systems. Both the harmful social media detection module and the traffic violation detection system showed that automated models can effectively identify threats, violations, and unsafe conditions in real time. This enhances scalability and allows systems to manage large amounts of data efficiently.

Additionally, the results highlight that predictive analytics plays a vital role in proactive safety management. The women and children safety prediction system showed it can identify high-risk areas and patterns, enabling preventive actions rather than just reactive ones. Similarly, traffic analysis helps spot accident-prone areas.

However, the study also pointed out some limitations. The performance of AI models depends on the availability of high-quality, localized datasets, especially for Sinhala-English mixed data. Furthermore, the system's real-time performance is affected by network connectivity, which may influence streaming and responsiveness in some environments.

Overall, the findings confirm that integrating AI, real-time data processing, and predictive modeling into a single platform can significantly enhance public safety, improve emergency response, and support data-driven decision-making in Sri Lanka.

4.3 Discussions

The results of the SafeLink system show that combining multiple AI-driven modules into one platform greatly improves public safety compared to traditional isolated systems. Each component plays a unique role, but their true strength comes from working together, allowing for both reactive and proactive safety management.

The Women and Children Safety Prediction module shows promise in identifying risk patterns using various types of data. The high accuracy in video detection (around 91.8%) confirms that computer vision models can reliably identify vulnerable individuals and related situations in real time. In contrast, the lower performance of the text classification model (about 71.8%) points to a limitation due to the lack of localized Sinhala-English datasets and the challenges of informal language. Still, combining video, audio, and text analysis enhances overall system reliability. This shows that using multiple types of data is more effective than relying on just one for safety prediction.

The Emergency SOS system marks a significant step forward from traditional voice-based emergency reporting methods. With real-time GPS tracking and live streaming, responders gain immediate situational awareness, reducing uncertainty during emergencies. The YOLOv11 model shows strong performance at a balanced confidence level (F1 approximately 0.69), indicating it can effectively detect visual threats like weapons and fire. Moreover, incorporating audio analysis using YAMNet further improves system reliability by identifying distress signals. However, the system's reliance on network connectivity could impact real-time performance in areas with poor coverage, which is an important limitation to note.

The Harmful Social Media Content Detection module effectively identifies explicit harmful content in multiple languages and code-mixed data. It can detect direct threats and violent language, showing that transformer-based models work well for understanding context. Adding computer vision improves detection accuracy by analyzing images and videos along with text. Nevertheless, challenges remain in spotting subtle or context-dependent forms of harmful content, such as sarcasm or implicit hate speech. This highlights the need for better contextual modeling and locally relevant datasets.

The Smart Traffic Violation Detection and Accident Prediction module performs well in detecting vehicles and license plates, achieving a strong precision rate (0.9449) and mAP scores (0.9714). These results confirm that YOLO-based object detection, combined with OCR techniques, is effective for real-world traffic monitoring. However, the lower performance in violation detection (mAP@0.5 to 0.95 about 0.5287) shows the challenges in interpreting contextual situations compared to recognizing objects directly. This suggests that we need to improve how we understand context and diversify datasets.

When we look at the system as a whole, integrating these four components creates a complete solution addressing various aspects of public safety. Unlike traditional systems that work separately, the SafeLink platform merges predictive analytics, real-time monitoring, and automated detection into one framework. This integration allows for valuable insights, such as combining risk prediction data with real-time emergency alerts to enhance prioritization and response efficiency. the system highlights the practical benefits of multimodal AI. By combining video, audio, text, and location data, it leads to more accurate and reliable decision-making. Using AI models across different areas reduces dependence on manual processes and boosts scalability, making the system suitable for real-world use.

However, several challenges remain. AI model performance depends on the availability and quality of local datasets, especially for multilingual and context-sensitive tasks. Real-time features like streaming and data transmission rely on network infrastructure, which might affect system performance in certain areas. Additionally, integrating multiple complex modules increases system complexity, requiring efficient backend design and resource management.

Overall, the results indicate that the SafeLink system is both feasible and effective in enhancing public safety through intelligent automation and real-time analysis. Combining multiple AI-driven components into a single platform is a significant move toward proactive, data-driven community policing in Sri Lanka. Future improvements can focus on enhancing dataset quality, optimizing model performance, and increasing system scalability to further strengthen the platform.

2. SUMMARY OF EACH STUDENT'S CONTRIBUTION

Table 5. 1 Individual Contribution

Student ID	Student Name	Individual Component	Contribution
IT22311054	Indudunu I W O	AI-Powered Women and Children Safety Risk Prediction System	25%
IT22003096	Amaradasa S A A M	Emergency SOS with Real-Time Location & Live Streaming	25%
IT22329660	Fernando P K S	Real-Time AI System for Detecting Harmful Social Media Content	25%
IT22331236	Jayakody J M P S B	Smart Traffic Violation Detection & Accident Prediction System	25%

3. CONCLUSION

This research presented the design and development of an AI-powered integrated public safety platform, SafeLink, aimed at improving emergency response, crime detection, and risk prediction in Sri Lanka. The system combines four key components: Smart Traffic Violation Detection and Accident Prediction, Real-Time Harmful Social Media Content Detection, Safety Prediction for Women and Children, and Emergency SOS with Real-Time Location and Live Streaming. By integrating these components into a single platform, the system addresses the limitations of existing solutions, which are mostly isolated, reactive, and lack real-time capabilities.

The proposed system utilizes advanced technologies such as computer vision, natural language processing, audio analysis, and machine learning to analyze data from multiple sources. The integration of multimodal data enables more accurate detection of incidents and better understanding of real-world situations. The Emergency SOS component enhances real-time communication between users and authorities, while predictive models support proactive decision-making by identifying high-risk areas and potential threats.

The implementation of the SafeLink system demonstrates that combining multiple safety domains into a unified platform can significantly improve the efficiency and effectiveness of public safety management. The system provides real-time alerts, predictive insights, and centralized monitoring through a user-friendly mobile application and administrative dashboard.

In conclusion, the SafeLink platform offers a practical and scalable solution to modern public safety challenges. Future improvements can include expanding dataset availability, improve model accuracy, and integrating the system with national-level infrastructure to further enhance its impact. Overall, this research contributes toward building safer and more resilient communities through the use of intelligent and integrated technologies.

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APPENDICES